MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : PEC-CS702E/PEC-IT702F Cyber Security

UPID : 007558

Time Allotted : 3 Hours *Full Marks :70*

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following : [1 x 10 = 10]

(I) What is the existence of weakness in a system or network is known as?

(II) Which DDoS in mobile systems wait for the owner to trigger the cyber attack?

(III) Why are the factors like Confidentiality, Integrity, Availability, and Authenticity considered as the fundamentals?

(IV) What refers to exploring the appropriate, ethical behaviors related to the online environment and digital media platform?

(V) What type of text is transformed with the help of a cipher algorithm?

(VI) Cyberspace" was coined by _____.

(VII) Network failure is primarily a _____ issue.

(VIII) _____ means the security of data from tampering by unidentified users.

(IX) Which is a sort of independent type of malicious program that would not require any host program?

(X) An act to injure, corrupt, or threaten a system or network is characterized_____.

(XI) What is security misconfiguration?

(XII) Differentiate Worm and Virus.

Group-B (Short Answer Type Question)

Answer *any three* of the following : [5 x 3 = 15]

2. Describe Security risk and analysis. [5]
3. Explain Data backup security, Data archival security, and Data disposal considerations. [5]
4. What is the use of policy? [5]
5. Define session hijacking and its various challenges. [5]
6. Explain the Feistel cipher in detail. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following : [15 x 3 = 45]

7. (a) Write short notes on: [5]
   Firewall.
   (b) Backdoor. [5]
   (c) Trapdoor. [5]
8. (a) Discuss the SQL Injection in detail. [5]
   (b) What is buffer overflow? [5]
   (c) Discuss how to minimize Buffer Overflow. [5]
9. (a) How VoIP hacking is done by attackers? [4]
   (b) What are the measures for it? [5]
   (c) Identify wireless network defense. [6]
10. (a) Name & explain the two protocols to use for E-mail Security. [8]
    (b) Explain Cryptanalysis process. [7]
11. (a) Elaborate "Cyber-crime plays a vital role against person, property, and govt. to protect all valuable information and rights". [6]
    (b) Encrypt a plain text using transposition cipher key. [6]
    (c) What is transposition cipher? [3]

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